

Understanding & Boost Consumer Behavior for Online Shopping

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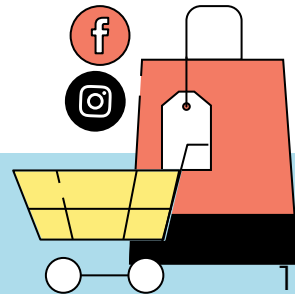
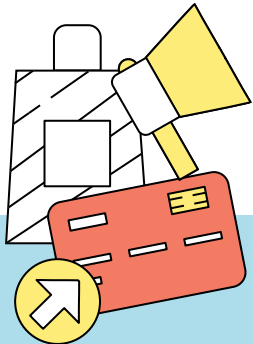


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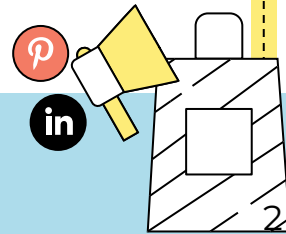
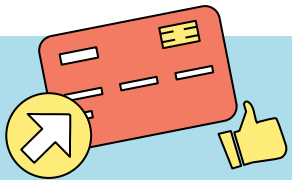
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**Analysis &
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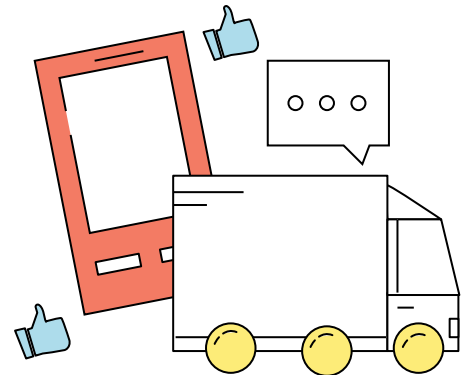
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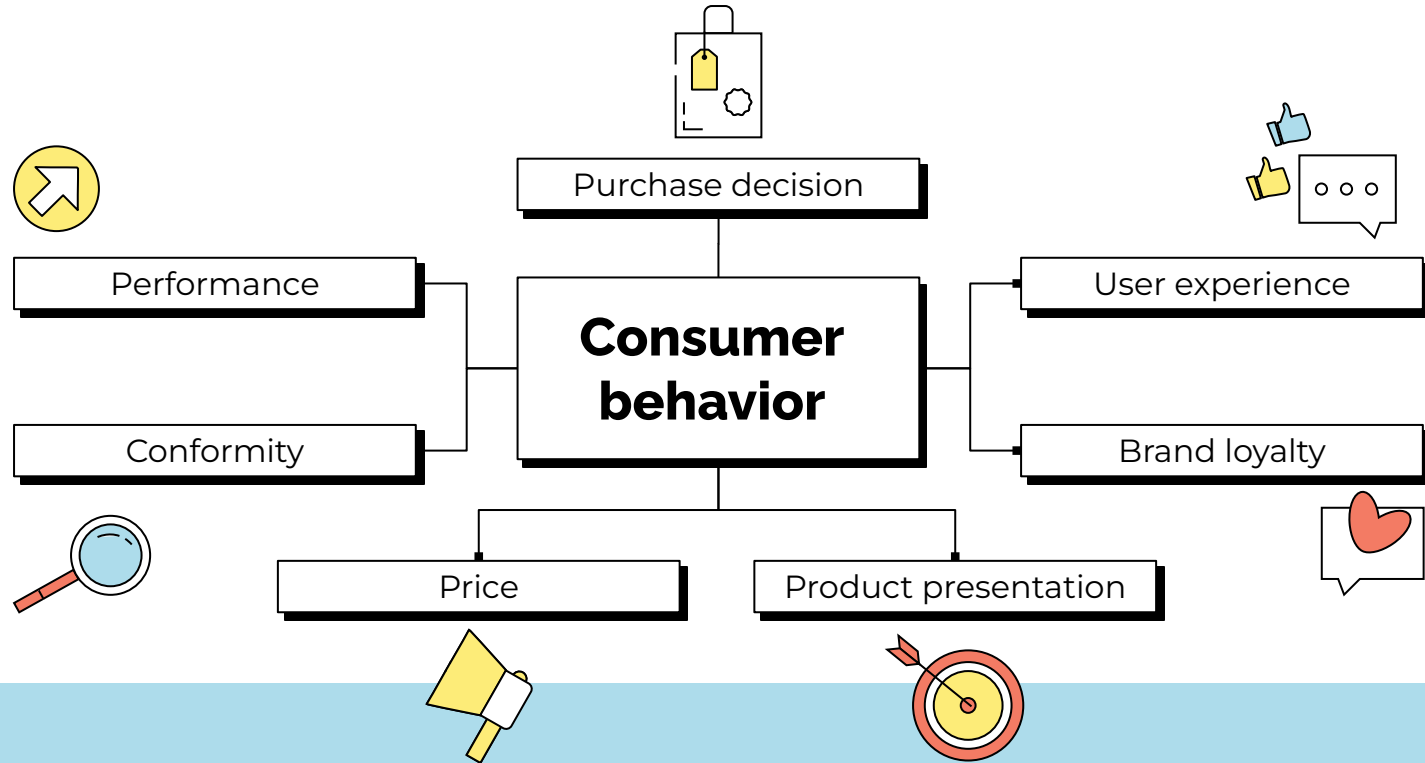


01

Introduction



Introduction



Online shopping is growing very fast. So, it is very important to understand how customers behave online

Objectives

Understanding

Understand customer behavior and engagement patterns by analyzing how customers browse and interact to identify what keeps them engaged.

Prediction

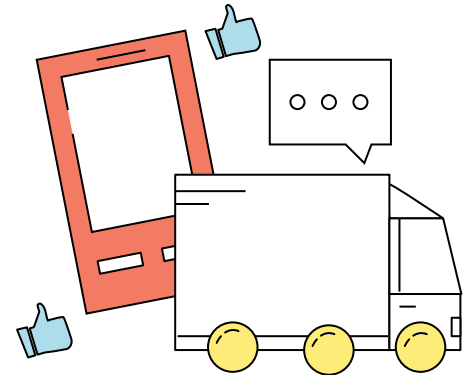
Use machine learning to predict and improve user engagement, improve the shopping experience and marketing effectiveness.

Clickstream Data for Online Shopping Dataset records clickstream activity from an online store selling clothing for pregnant women over five months in 2008.

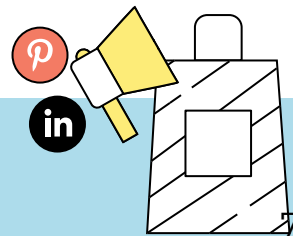
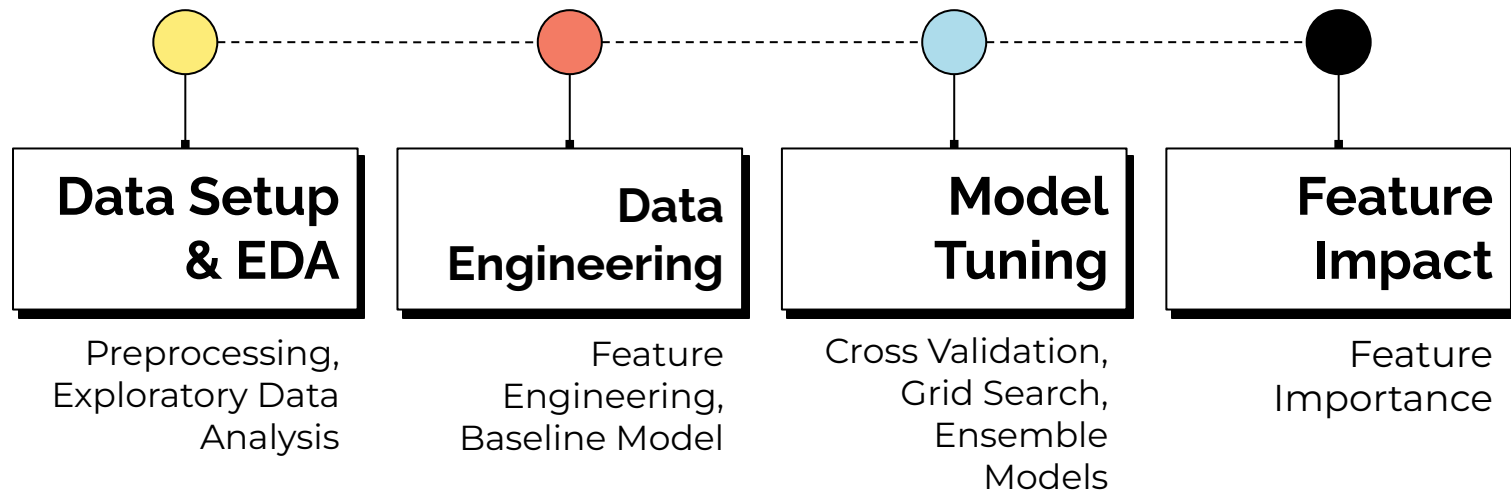


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Methodology

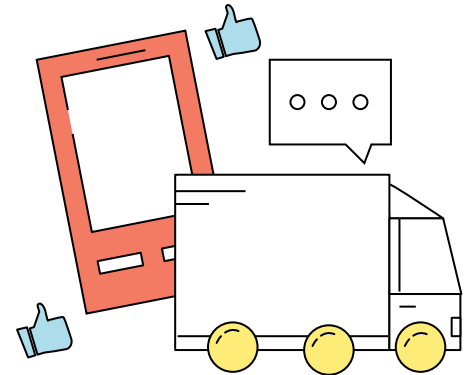


Methodology



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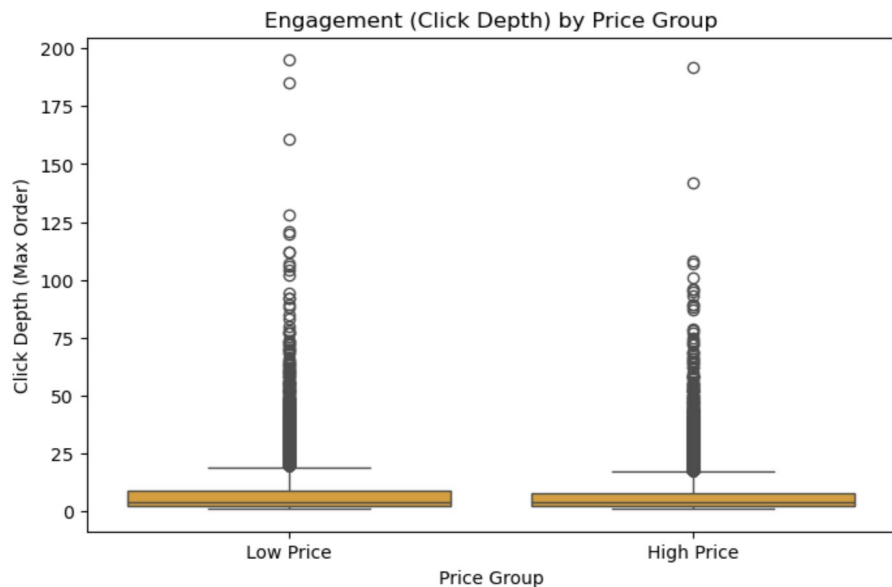
Analysis & Insights





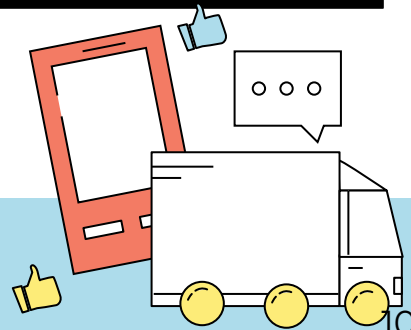
Step 1: Data Setup & EDA

Product Price Influences

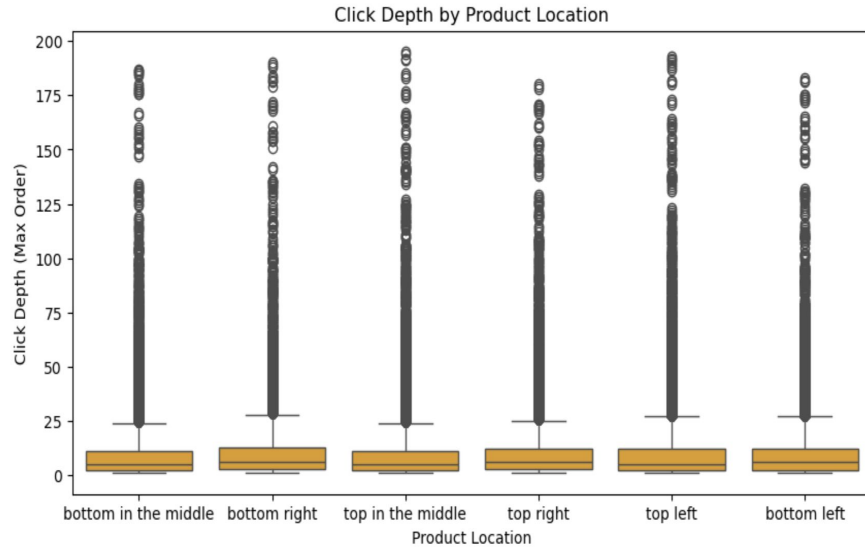


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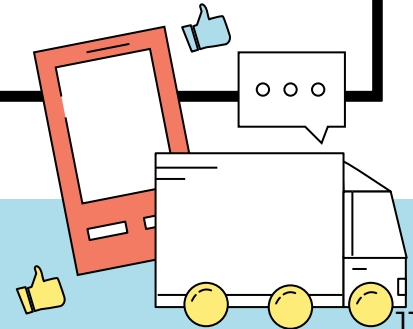
- Sessions with **lower-priced products show deeper click behavior** (avg. 7.0 clicks (low price) vs 6.7 clicks (high price))
- The difference is **small but statistically significant**



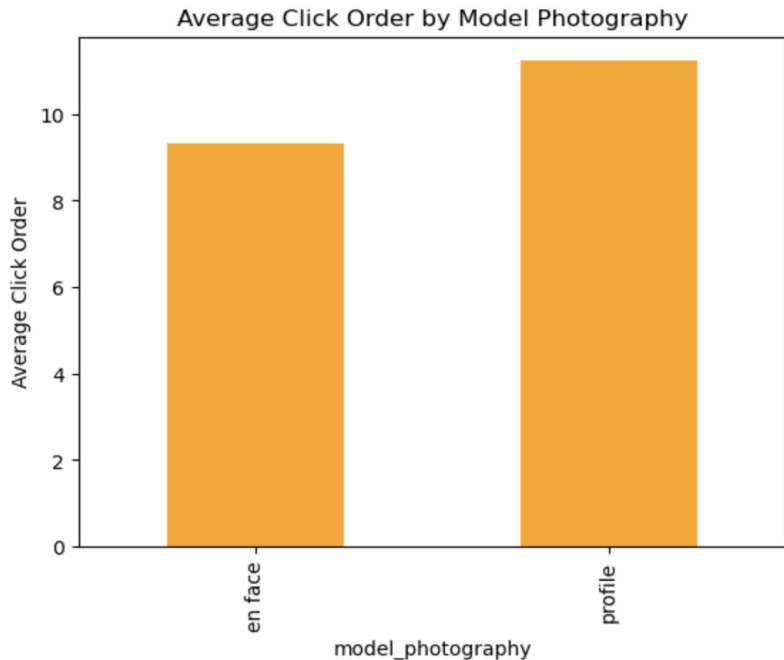
Product Presentation Affects Engagement



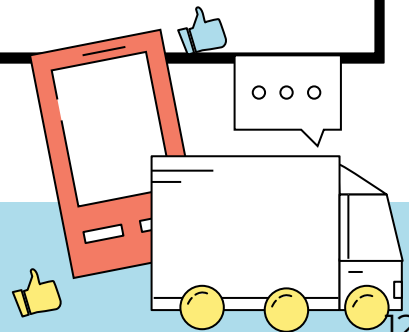
- Top middle and top left (195 clicks) **show slightly higher engagement**, suggesting that visual placement is important in online shopping



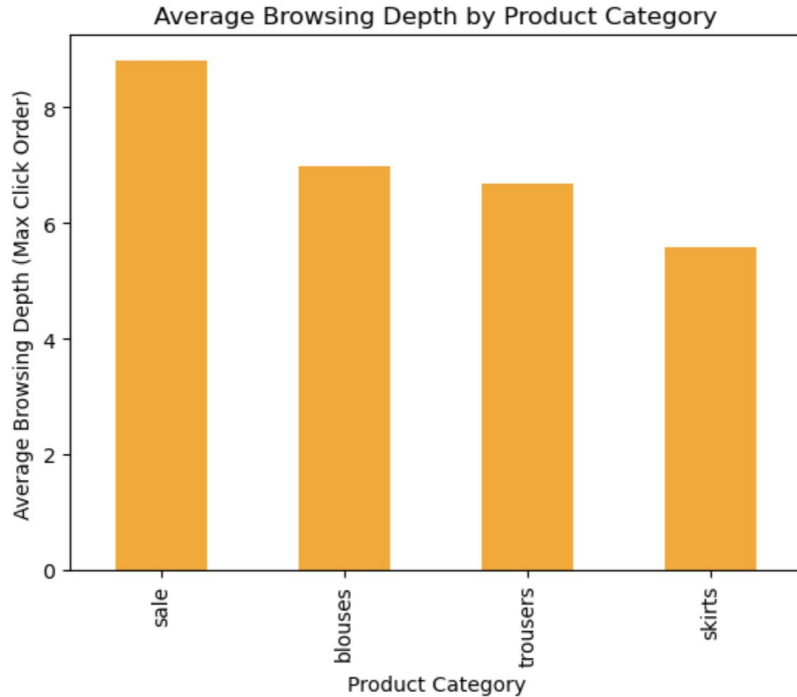
Model Photography Influences Early Clicks



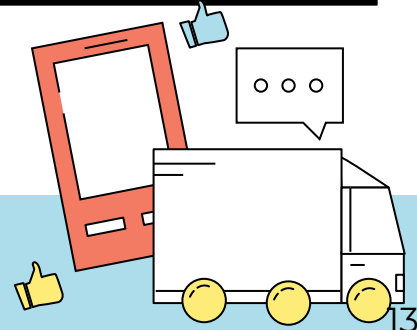
- En face images have a lower mean click order, meaning **en face images attracting clicks earlier** in the session
- Model photography style influences initial user engagement



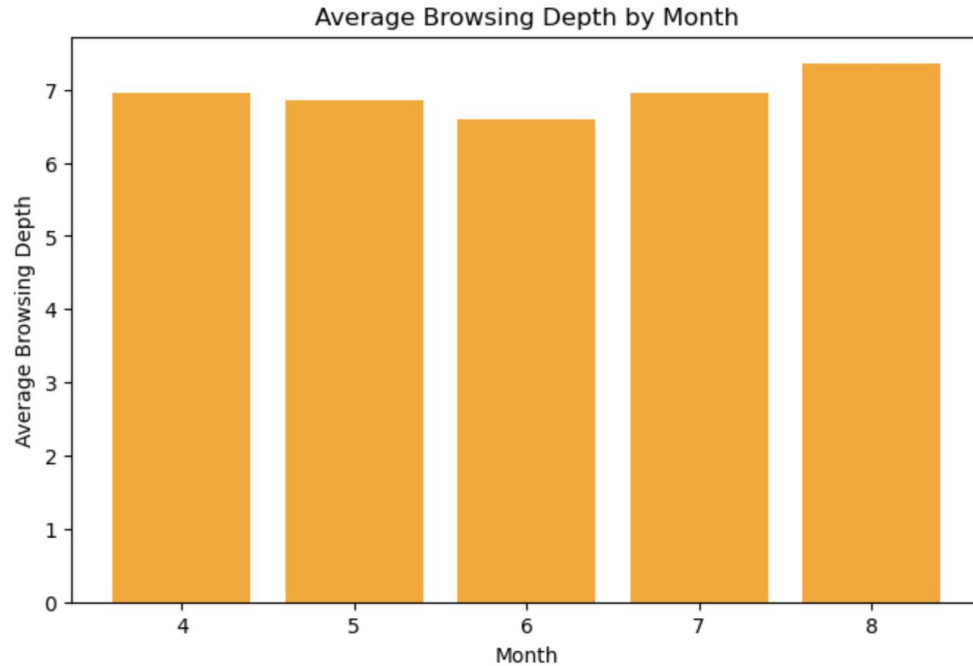
Category Drives Browsing Depth



- Sale items drive the deepest engagement (avg. 8.8 clicks)
- Skirts show shallow browsing (avg. 5.6 clicks)

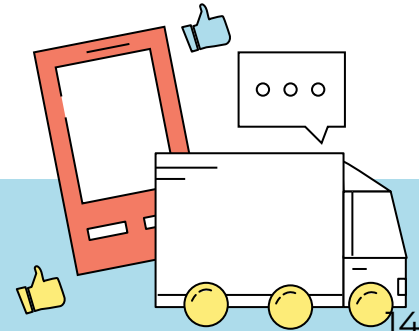


Seasonality Affects Browsing



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- Engagement is stable during most of the year
- Highest browsing depth in August



Step 2: Data Engineering



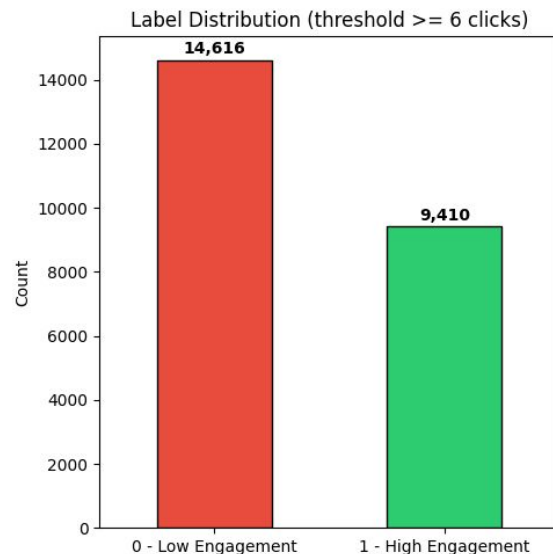
Feature Engineering

Feature Engineering

- Convert the click-based dataset into a **session-based dataset** by grouping events using the session ID
- **Create additional features** to better represent user behavior within a session (e.g., 'dominant_category', 'dominant_colour',...)
- Transform features:
 - Categorical features using One-Hot Encoding
 - Numerical features using Standard Scaling

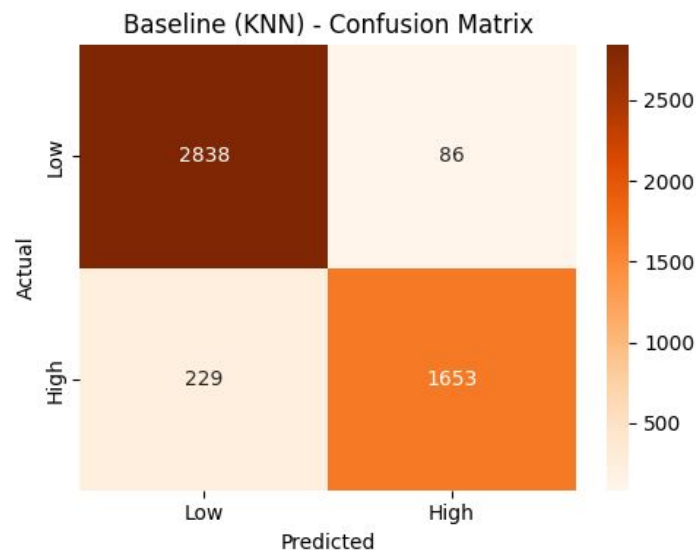
Label Creation

- Create a **binary engagement label** based on the total number of clicks per session
 - **Label 0** (low engagement): sessions with few clicks
 - **Label 1** (high engagement): sessions with many clicks (top 40%)
- Define engagement threshold (6 clicks) using click distribution

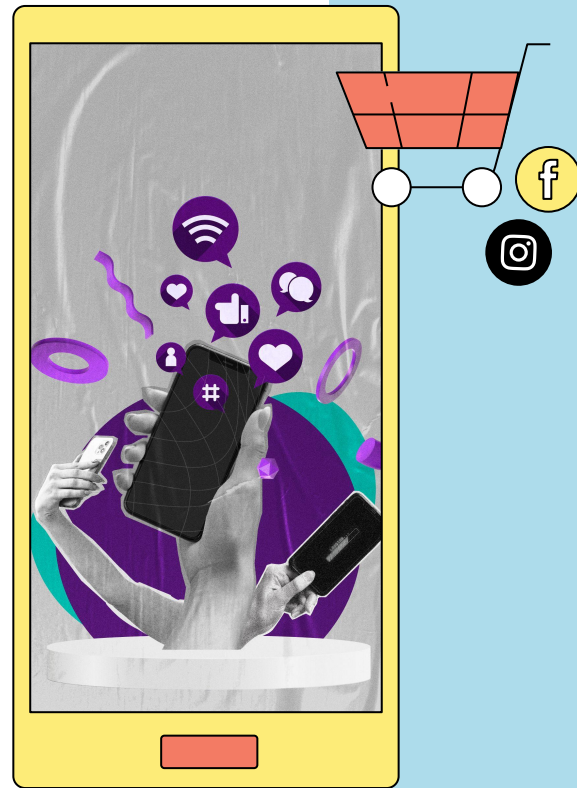
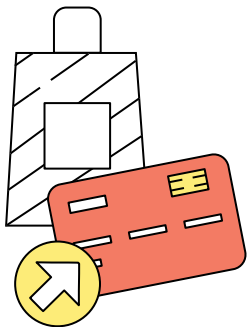


Baseline Model: KNN

- **Data split:** 80% training / 20% testing
- **Baseline model:** k-Nearest Neighbors (KNN)
- **Performance:** accuracy 93%
- KNN provides a **strong baseline**, especially for identifying low-engagement sessions, with room to improve recall on high engagement.

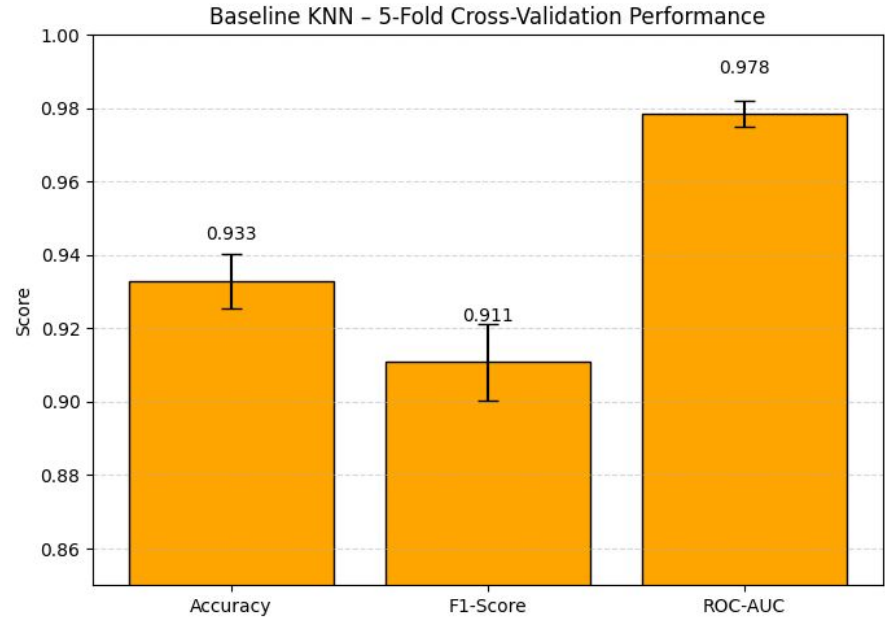


Step 3: Model Tuning

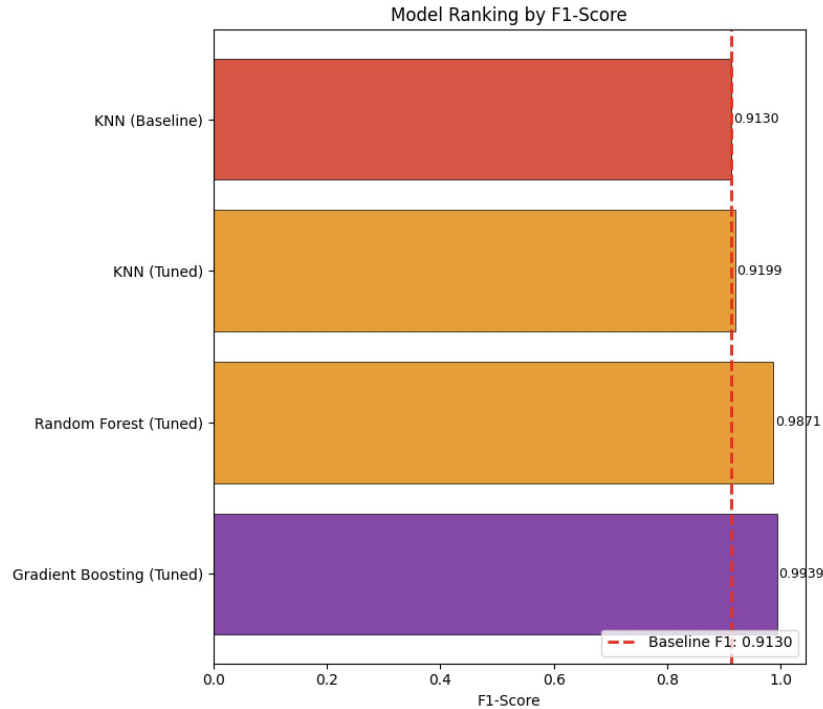


Cross-Validation

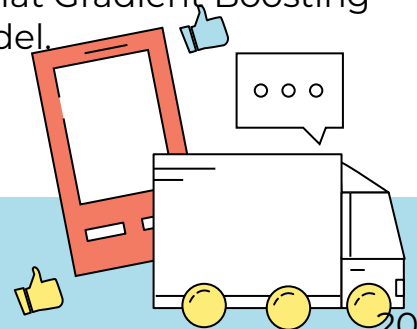
- **Method:** 5-fold cross-validation
- **Model:** KNN (baseline)
- **Results**
 - Achieves 93.3% accuracy and F1-score of 91.1%, indicating balanced performance across classes
 - The model is stable and generalizes well to unseen data
- Confirms KNN as a **reliable** and **solid baseline** for further model comparison



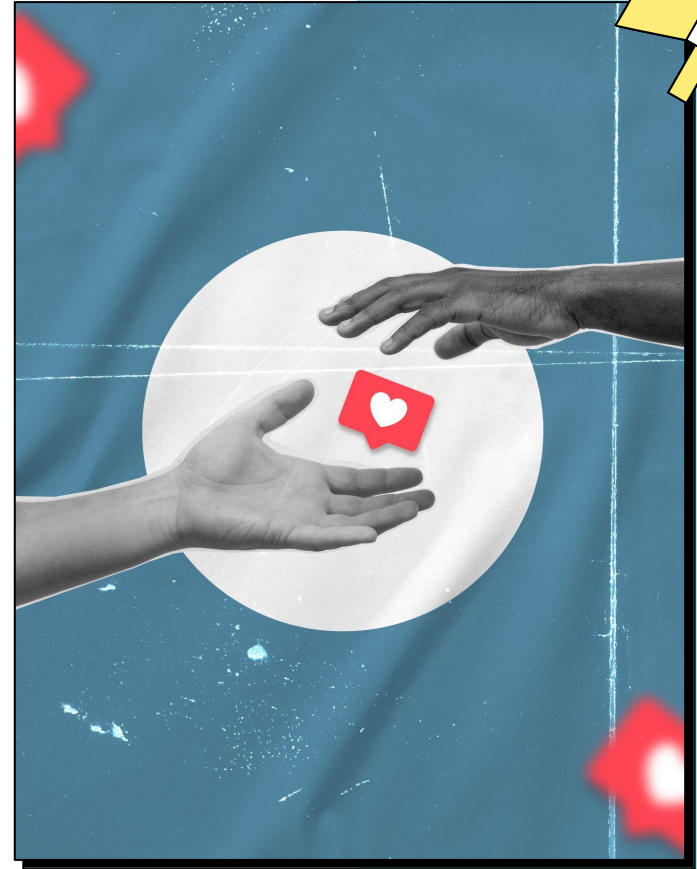
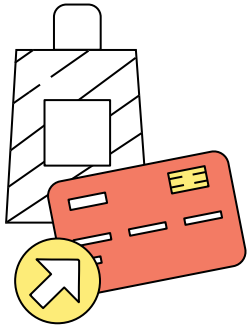
Hyperparameters Tuning



- **Method:** 5-fold Stratified Cross-Validation using GridSearchCV
- **Models tuned:** 3 classification models
- **Results**
 - Optimal hyperparameters identified for each model
 - **Best-performing model: Fine-tuned Gradient Boosting**
 - Achieved **99.4% accuracy**, significantly outperforming the baseline
- **Hyperparameter tuning greatly improves performance** and shows that Gradient Boosting is the best-performing model.



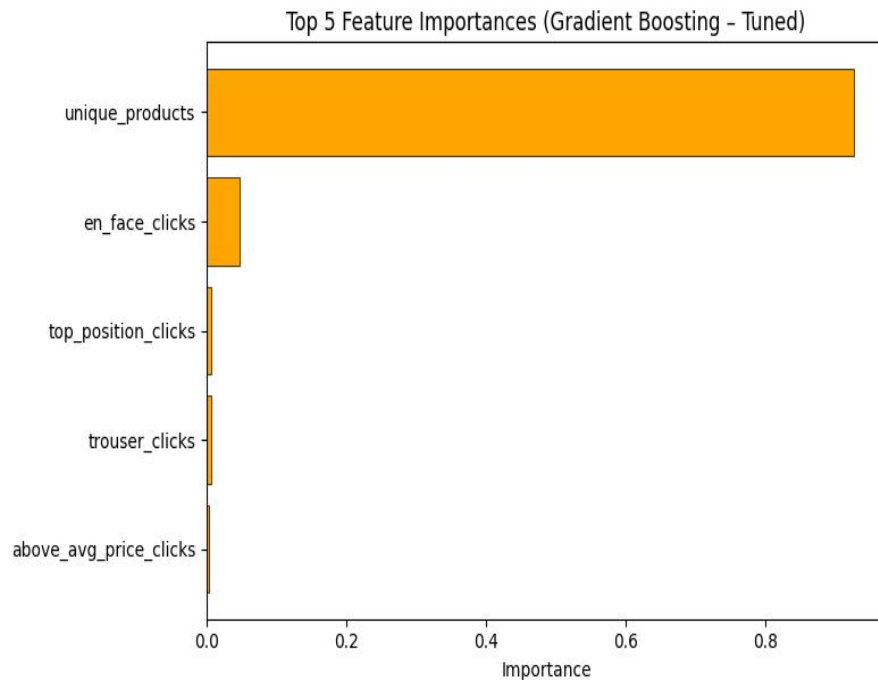
Step 4: Feature Impact



Feature Importance

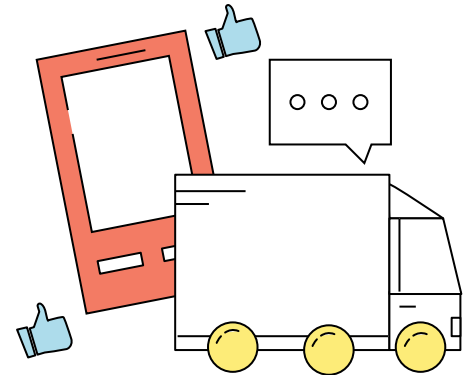
Extract feature importance from the best fine-tuned Gradient Boosting model, based on how much each feature reduces prediction error across all trees

- *'unique_products'* (~93%) is the **most influential feature**, indicating that the diversity of products a user interacts with strongly drives predictions
- *'en_face_clicks'* (~4.7%), capturing **user engagement with face-category products**
- Other features (<1%), such as *position_clicks* or category-specific clicks have minimal individual impact, suggesting the model relies more on overall product diversity than fine-grained click patterns



04

Conclusion



Conclusion

- **User engagement in online shopping can be predicted** effectively using machine learning
- **Price, product position, image style, category, and seasonality** influence browsing behavior
- **Product diversity viewed is the strongest factor** for high engagement
- Insights can help improve:
 - Product recommendations
 - Page layout and design
 - Marketing strategies
- **Data-driven decisions can improve the online shopping experience**

Resources

- Github Final Project
- Clickstream Data for Online Shopping Dataset
- Slide Templates



Thanks!



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